

VITAMINS

Definition, Classification, Biochemical Functions and Deficiency Disorders

High-yield exam focus

- Vitamins are required in small amounts, but their deficiency produces characteristic clinical disorders.
- Fat-soluble vitamins mainly depend on fat absorption and can accumulate; water-soluble vitamins usually need regular dietary supply.
- Most B-complex vitamins work as coenzymes in energy metabolism, amino acid metabolism, one-carbon transfer or blood cell formation.

1. Definition of Vitamins

Vitamins are organic compounds required in the diet in small amounts for specific biological functions needed for normal growth, maintenance of health and metabolism. They do not directly supply energy like carbohydrates or fats, but many of them act as coenzymes or hormone-like regulators.

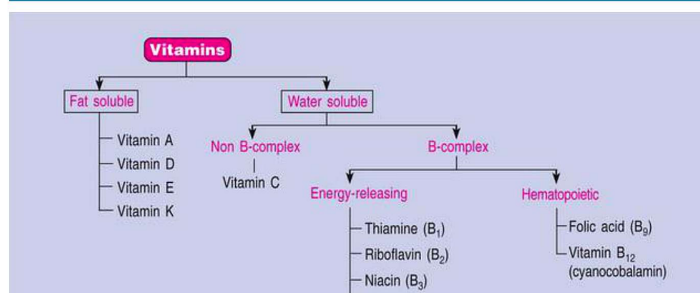
The body cannot synthesize most vitamins in adequate quantity. Therefore, they must be supplied through diet. Their deficiency causes specific biochemical defects that later appear as clinical symptoms.

2. Classification of Vitamins

Vitamins are mainly classified on the basis of solubility:

Class	Examples	Important features
Fat-soluble vitamins	A, D, E, K	Absorbed with dietary fat; bile salts are needed; stored in liver/adipose tissue; deficiency appears slowly; toxicity is more likely with excess intake.
Water-soluble vitamins	B-complex and vitamin C	Dissolve in water; most are not stored in large amounts; excess is usually excreted in urine; deficiency may appear faster with poor intake.

Table 7.1
Classification of vitamins



Source diagram: Classification of vitamins into fat-soluble and water-soluble groups.

3. Biochemical Functions and Disorders - Quick Table

Vitamin	Active form / coenzyme	Main biochemical function	Deficiency disorder
A	Retinol, retinal, retinoic acid; beta-carotene as provitamin	Vision, epithelial maintenance, growth, immunity, reproduction.	Night blindness, xerophthalmia, Bitot spots, keratomalacia, dry skin, infection tendency.
D	Calcitriol (1,25-dihydroxycholecalciferol)	Calcium and phosphate absorption, bone mineralization, calcium	Rickets in children, osteomalacia in adults, hypocalcemic tetany;

Vitamin	Active form / coenzyme	Main biochemical function	Deficiency disorder
		homeostasis.	excess causes hypercalcemia.
E	Tocopherols/tocotrienols	Antioxidant protection of membranes; protects PUFA and RBCs from oxidative damage.	Hemolytic anemia, neuromuscular problems; reproductive failure in many animals.
K	Phylloquinone, menaquinones	Gamma-carboxylation of clotting factors II, VII, IX, X and proteins C/S.	Bleeding tendency, prolonged prothrombin time, hemorrhagic disease of newborn.
C	Ascorbic acid	Collagen hydroxylation, antioxidant action, iron absorption, wound healing, immunity.	Scurvy: bleeding gums, loose teeth, poor wound healing, petechiae, anemia.
B1	Thiamine pyrophosphate (TPP)	Oxidative decarboxylation and transketolase reactions; nerve metabolism.	Beriberi, Wernicke-Korsakoff syndrome.
B2	FMN, FAD	Oxidation-reduction reactions in energy metabolism.	Cheilosis, angular stomatitis, glossitis, dermatitis.
B3	NAD ⁺ , NADP ⁺	Redox reactions in carbohydrate, lipid and amino acid metabolism.	Pellagra: dermatitis, diarrhea, dementia; may lead to death.
B6	Pyridoxal phosphate (PLP)	Amino acid metabolism, neurotransmitter synthesis, heme synthesis.	Peripheral neuropathy, convulsions, sideroblastic anemia, dermatitis.
Biotin	Biocytin/carboxybiotin form	Carboxylation reactions in gluconeogenesis, fatty acid synthesis and amino acid metabolism.	Dermatitis, alopecia, enteritis; raw egg white can cause deficiency.
Pantothenic acid	Coenzyme A, acyl carrier protein	Acyl group transfer; central role in carbohydrate, lipid and amino acid metabolism.	Rare; fatigue, paresthesia, burning feet syndrome, GI symptoms.
Folic acid	Tetrahydrofolate (THF)	One-carbon transfer; purine, dTMP and RBC formation.	Megaloblastic anemia, glossitis, neural tube defects, hyperhomocysteinemia.
B12	Methylcobalamin, deoxyadenosylcobalamin	Methionine synthase and methylmalonyl-CoA mutase; RBC and nerve function.	Pernicious/megaloblastic anemia, neurological degeneration, methylmalonic aciduria.

4. Fat-Soluble Vitamins

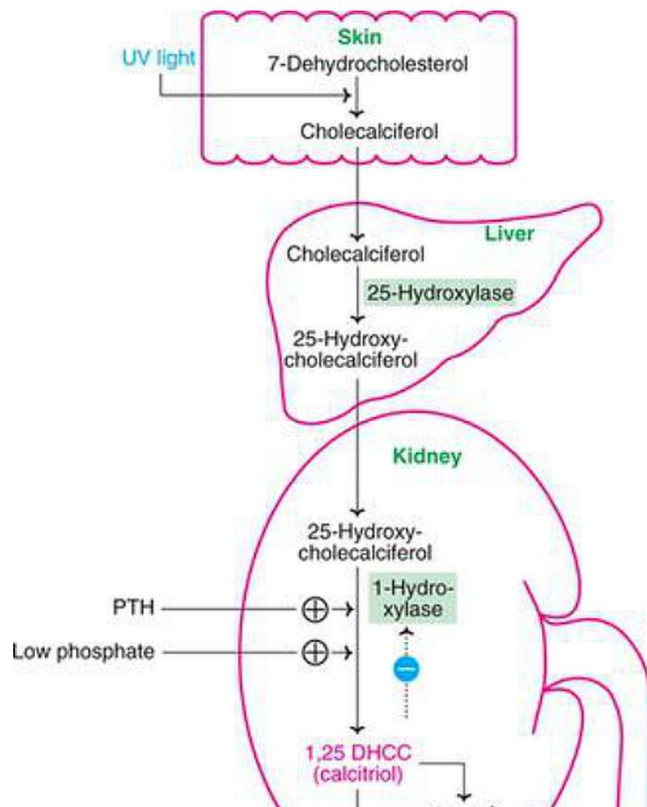
Vitamin A (Retinoids)

- Important forms: retinol, retinal and retinoic acid; beta-carotene is a provitamin from plant sources.
- Vision: 11-cis retinal combines with opsin to form rhodopsin, the visual pigment required for dim-light vision.
- Epithelial function: maintains normal epithelial differentiation and mucus secretion; deficiency causes keratinization and dryness.
- Growth, reproduction and immunity: retinoids regulate gene expression and support immune defense.
- Deficiency: night blindness appears early; severe deficiency causes xerophthalmia, Bitot spots, keratomalacia and blindness. Skin becomes dry and rough; infection tendency increases.
- Toxicity: excessive intake may cause headache, dermatitis, raised intracranial tension, liver enlargement and bone pain; high intake during pregnancy is harmful.

Vitamin D (Calciferol)

- Vitamin D2 is ergocalciferol; vitamin D3 is cholecalciferol. The biologically active form is calcitriol: 1,25-dihydroxycholecalciferol.

- Activation: vitamin D is hydroxylated first in liver to 25-hydroxycholecalciferol and then in kidney to calcitriol.
- Biochemical function: calcitriol increases intestinal absorption of calcium and phosphate, supports bone mineralization and helps maintain plasma calcium.
- Deficiency: rickets in children, osteomalacia in adults, bone pain, deformities and hypocalcemic tetany.
- Toxicity: hypervitaminosis D causes hypercalcemia, soft-tissue calcification and renal damage/stones.



Source diagram: Metabolism and biochemical functions of vitamin D.

Vitamin E (Tocopherols)

- Vitamin E is a lipid-soluble antioxidant, especially important in biological membranes.
- It protects polyunsaturated fatty acids and membrane lipids from free-radical-mediated peroxidation.
- It helps preserve RBC membrane stability and supports normal neuromuscular function.
- Deficiency is uncommon but may occur in fat malabsorption or premature infants. It may cause hemolytic anemia, myopathy, neuropathy and ataxia.
- Very high doses can antagonize vitamin K action and increase bleeding tendency.

Vitamin K

- Forms: vitamin K1 (phylloquinone) from plants, vitamin K2 (menaquinones) from intestinal bacteria and vitamin K3 (menadione, synthetic).
- Main biochemical function: vitamin K is required for gamma-carboxylation of glutamate residues in clotting factors II, VII, IX and X. This enables calcium binding and activation of clotting proteins.
- Deficiency: prolonged clotting time, hemorrhage and hemorrhagic disease of newborn. It may occur after fat malabsorption, obstructive jaundice or prolonged broad-spectrum antibiotic therapy.
- Antagonists: dicumarol and warfarin inhibit vitamin K recycling and are used as anticoagulants.

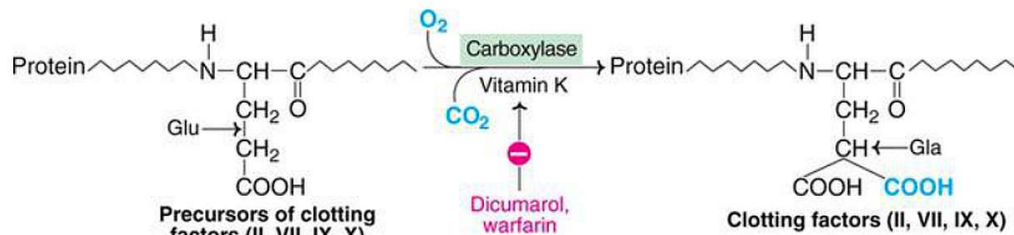
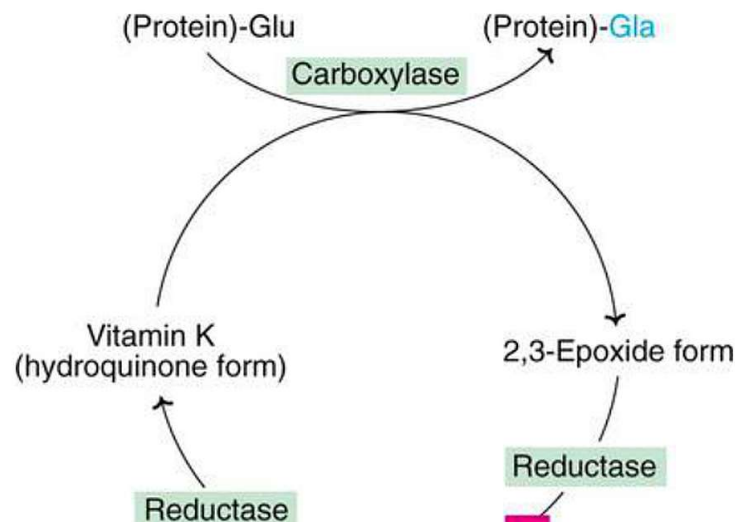


FIG. 7.12 Vitamin K dependent carboxylation of the precursors of clotting factors.

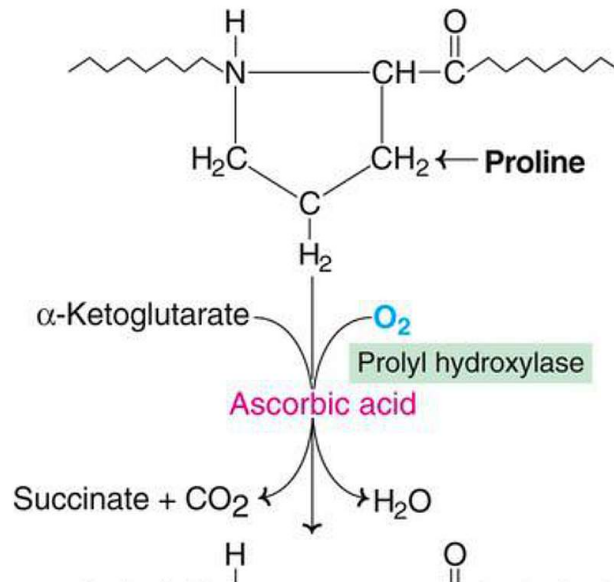


Source diagram: Vitamin K dependent carboxylation and vitamin K cycle.

5. Water-Soluble Vitamins

Vitamin C (Ascorbic Acid)

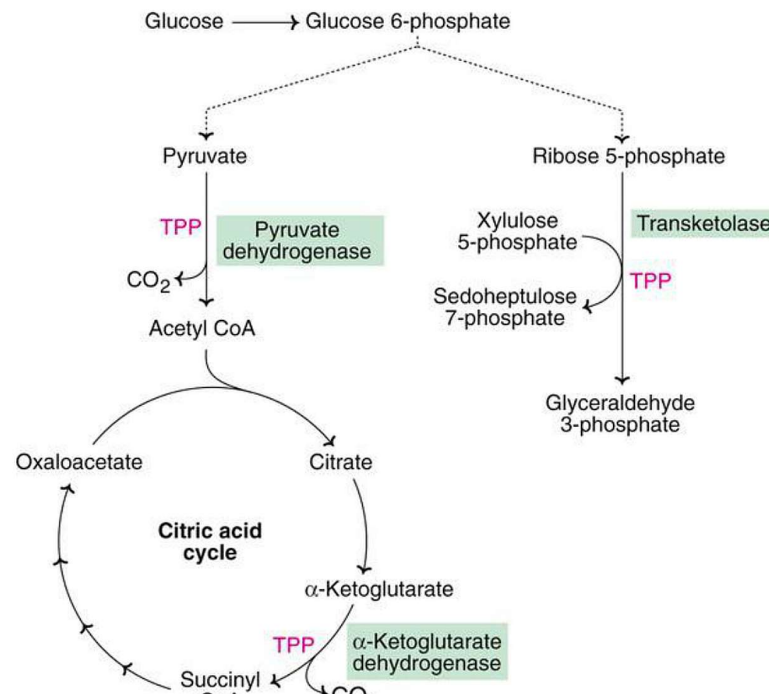
- Vitamin C functions mainly as a reducing agent and antioxidant.
- It is required for hydroxylation of proline and lysine during collagen synthesis. This explains its role in wound healing, connective tissue strength and capillary integrity.
- It increases iron absorption by maintaining iron in the ferrous state and helps in folate metabolism, immune function and hormone synthesis.
- Deficiency causes scurvy: swollen/bleeding gums, loose teeth, petechiae, poor wound healing, anemia, bone pain and reduced resistance to infection.
- Very high doses may cause gastrointestinal irritation and increase oxalate load in susceptible persons.



Source diagram: Ascorbic acid dependent hydroxylation of proline in collagen formation.

Vitamin B1 - Thiamine

- Active coenzyme: thiamine pyrophosphate (TPP).
- TPP is required for oxidative decarboxylation of pyruvate, alpha-ketoglutarate and branched-chain ketoacids; it also participates in transketolase reaction of HMP shunt.
- Deficiency affects tissues with high energy requirement, especially nerves and heart.
- Disorders: dry beriberi (peripheral neuropathy), wet beriberi (cardiac failure and edema), infantile beriberi and Wernicke-Korsakoff syndrome in alcoholism.



Source diagram: Important thiamine/TPP-dependent reactions.

Vitamin B2 - Riboflavin

- Active coenzymes: flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD).

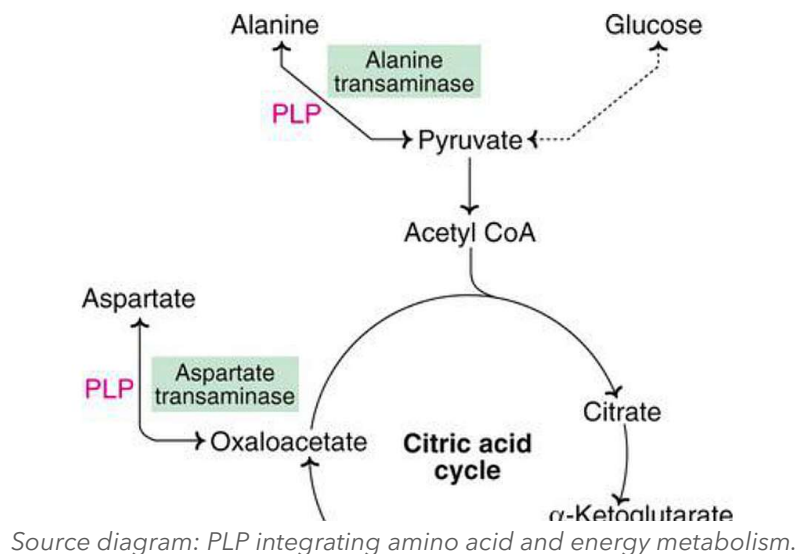
- FMN and FAD participate in oxidation-reduction reactions by accepting and donating hydrogen/electrons. They are important in energy-producing pathways and electron transport.
- Deficiency: cheilosis, angular stomatitis, glossitis, seborrheic dermatitis and corneal vascularization.

Vitamin B3 - Niacin

- Active coenzymes: NAD⁺ and NADP⁺.
- NAD⁺ mainly participates in catabolic oxidation reactions that generate energy; NADPH is important in reductive biosynthesis and antioxidant defense.
- Niacin may be obtained from diet or synthesized from tryptophan.
- Deficiency causes pellagra, classically remembered as the 3 Ds: dermatitis, diarrhea and dementia; untreated severe disease may lead to death.
- Excess nicotinic acid may cause flushing, itching, hepatotoxicity and hyperuricemia.

Vitamin B6 - Pyridoxine

- Active coenzyme: pyridoxal phosphate (PLP).
- PLP is central in amino acid metabolism: transamination, decarboxylation, deamination and condensation reactions.
- It is needed for neurotransmitter synthesis, heme synthesis and glycogen phosphorylase activity.
- Deficiency: peripheral neuropathy, convulsions, sideroblastic anemia, dermatitis, cheilosis and glossitis. Isoniazid therapy can produce B6 deficiency unless supplemented.



Biotin

- Biotin acts as a coenzyme for carboxylation reactions where carbon dioxide is added to a substrate.
- Important biotin-dependent enzymes include pyruvate carboxylase, acetyl-CoA carboxylase and propionyl-CoA carboxylase.
- Functions: gluconeogenesis, fatty acid synthesis and metabolism of certain amino acids/odd-chain fatty acids.
- Deficiency is rare. Raw egg white contains avidin, which binds biotin and prevents its absorption. Deficiency may cause dermatitis, alopecia, enteritis, depression and neurological symptoms.

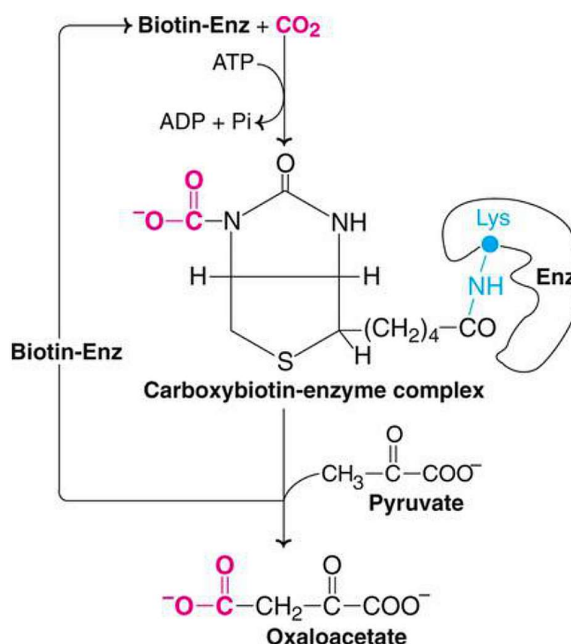


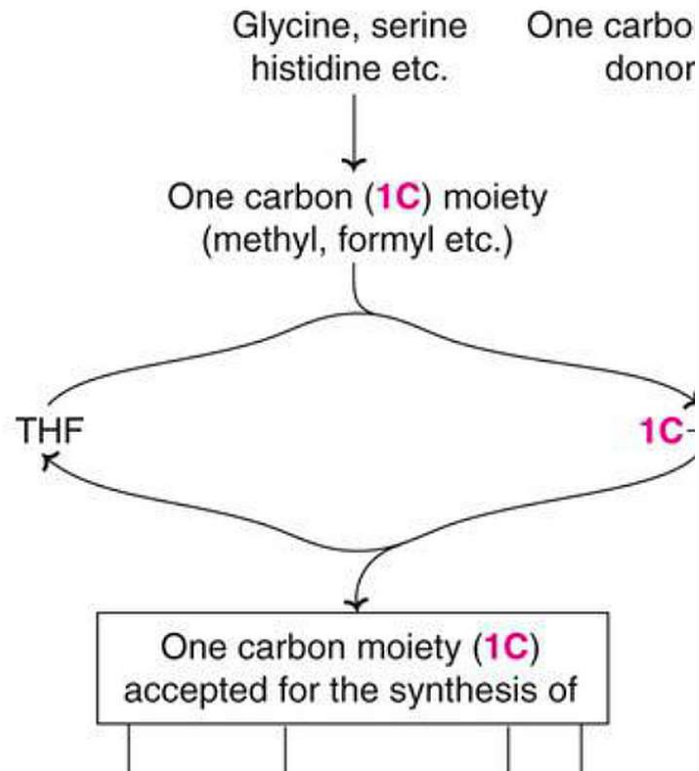
FIG. 7.29 Role of biotin in the carboxylation reaction, catalysed by the
Source diagram: Biotin-dependent carboxylation reaction.

Pantothenic Acid

- Active forms: coenzyme A (CoA) and acyl carrier protein.
- CoA carries activated acyl groups as thioesters. It is therefore essential in carbohydrate oxidation, beta-oxidation, fatty acid synthesis, cholesterol/steroid synthesis and amino acid metabolism.
- Deficiency is rare due to wide distribution in foods. Symptoms may include fatigue, irritability, numbness, paresthesia, gastrointestinal disturbance and burning feet syndrome.

Folic Acid

- Active coenzyme: tetrahydrofolate (THF).
- THF carries one-carbon units such as methyl, methylene and formyl groups.
- Functions: synthesis of purines, thymidylate (dTMP), methionine formation and maturation of RBCs.
- Deficiency: megaloblastic anemia, glossitis, weakness and hyperhomocysteinemia. During pregnancy, deficiency increases risk of neural tube defects in the fetus.
- Antagonists such as methotrexate inhibit folate metabolism and are used in selected therapeutic settings.



Source diagram: Overview of one-carbon metabolism involving THF.

Vitamin B12 - Cobalamin

- Active coenzymes: methylcobalamin and 5-deoxyadenosylcobalamin.
- Two important human reactions require B12: conversion of homocysteine to methionine and conversion of methylmalonyl-CoA to succinyl-CoA.
- B12 is essential for DNA synthesis, RBC maturation and maintenance of nervous tissue. Absorption needs intrinsic factor from the stomach.
- Deficiency: pernicious anemia/megaloblastic anemia, glossitis and neurological changes such as peripheral neuropathy and subacute combined degeneration. Methylmalonic acid may increase.
- Causes include intrinsic factor deficiency, gastrectomy, malabsorption and strict vegetarian diet without supplementation.

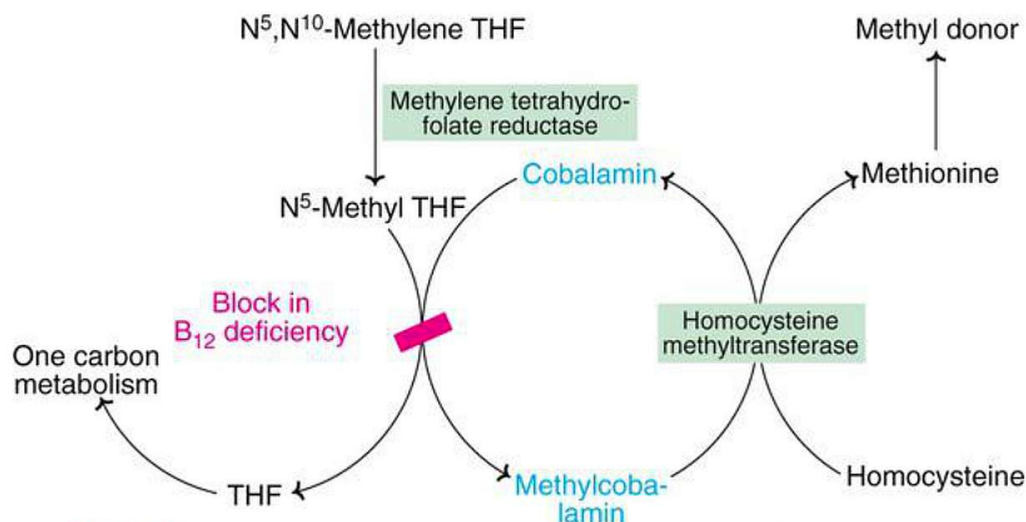


FIG. 7.39 Interrelationship between folic acid and vitamin B₁₂.

Source diagram: Interrelationship between folic acid and vitamin B₁₂; methyl trap concept.

6. Disorder-Oriented Revision List

Disorder	Vitamin involved	Key reason
Night blindness	Vitamin A	Early sign due to defective rhodopsin cycle.
Xerophthalmia / keratomalacia	Vitamin A	Dryness and damage to conjunctiva/cornea; can cause blindness.
Rickets	Vitamin D	Poor mineralization of growing bone in children.
Osteomalacia	Vitamin D	Defective mineralization of adult bone.
Bleeding tendency / prolonged PT	Vitamin K	Defective activation of clotting factors.
Scurvy	Vitamin C	Defective collagen formation; bleeding gums and poor wound healing.
Beriberi	Vitamin B ₁	Peripheral neuropathy and/or cardiac failure due to defective energy metabolism.
Pellagra	Vitamin B ₃	Dermatitis, diarrhea and dementia.
Sideroblastic anemia / neuropathy	Vitamin B ₆	Disturbed heme synthesis and amino acid/neurotransmitter metabolism.
Megaloblastic anemia	Folate or B ₁₂	Impaired DNA synthesis in rapidly dividing cells.
Neural tube defects	Folate	Risk increases with folate deficiency during early pregnancy.
Pernicious anemia with neurologic signs	Vitamin B ₁₂	Intrinsic factor deficiency or malabsorption; neurologic signs help distinguish B ₁₂ from folate deficiency.

Fast memory hooks

- A = eye and epithelium.
- D = calcium, phosphate and bone.
- E = antioxidant for membranes.
- K = koagulation factor activation.
- C = collagen and scurvy.
- B₁ = beriberi; B₃ = pellagra; B₉/B₁₂ = megaloblastic anemia; B₁₂ also affects nerves.

Source basis: Prepared from the Vitamins chapter of the uploaded Biochemistry textbook by U. Satyanarayana and U. Chakrapani. Figures are taken from the provided source PDF where shown.